# Free Will and Framework Inertia

A Textbook-Oriented Draft for the `gpt-mock` Workspace

LLM Mock Notes

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**Abstract**

This draft upgrades the earlier short note into a fuller textbook prototype. The goal is not to claim final rigor, but to create a manuscript structure that can support systematic teaching: a formal language, a chapter sequence, worked cases, intervention patterns, exercises, and reusable planning templates. The central claim remains unchanged: free will is most effective when it is deployed at low-inertia transition points and reinforced by pre-committed structure.

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# Chapter 1

# Why This Text Exists

The original manuscript established a useful thesis: people often fail to initiate beneficial action not because they lack values, but because they are embedded in a current behavioural frame whose reward structure, cognitive momentum, and environment all resist transition. That thesis is still the center of the project. What changes in this draft is the teaching ambition.

This version is written as a textbook prototype rather than a short conceptual essay. A textbook needs more than correct intuition. It needs ordered definitions, stable notation, chapter progression, worked cases, and an exercise layer that lets the reader test whether the framework actually explains behaviour rather than merely sounding insightful.

## 1.1 Learning Objectives

After reading this draft, a reader should be able to:

1. describe a behavioural state as a structured object rather than a mood;
2. distinguish reward, switching cost, and environmental reinforcement;
3. identify decision windows at which free will has unusually high leverage;
4. explain why planning changes the geometry of later action rather than merely adding motivation;
5. classify real-life failures using a reusable case vocabulary;
6. design interventions that lower activation barriers before failure occurs.

## 1.2 Book Architecture

The manuscript is organized around six teaching jobs:

1. Chapters 2 and 3 define the language of the model.
2. Chapter 4 explains why planning is structurally necessary.
3. Chapter 5 makes the model concrete across multiple domains.
4. Chapter 6 turns analysis into design patterns.
5. Chapter 7 tests whether the reader can reuse the framework.

## 1.3 Scope and Limits

This is still a semi-formal project. The purpose is to create a strong bridge between mathematical style and behavioural usefulness. Therefore:

- the notation is disciplined but intentionally lightweight;

- the propositions are explanatory tools, not claims of finished empirical law;
- the case library is designed to improve transfer, not to replace psychological literature;
- the intervention patterns are production templates for later expansion.

**Chapter Summary**

This chapter changes the business value of the manuscript. The project is no longer just a single argument. It is now a reusable teaching system with a defined learning path, explicit scope, and a structure that can absorb future proofs, cases, and exercises without collapsing into notes.

# Chapter 2

# Formal Language of Behavioural States

## 2.1 State, Frame, and Resistance

**Definition 2.1** (Behavioural state)**.** A behavioural state $s \in \mathcal{S}$ is a sustained mode of action together with its associated posture, attention pattern, local environment, and reward expectation.

**Definition 2.2** (Framework inertia)**.** The framework inertia of a state $s$ is

$$\mathrm{FI}(s) = \alpha \mathrm{R}(s) + \beta \mathrm{C}(s) + \gamma \mathrm{E}(s),$$

where $\mathrm{R}(s)$ is the immediate reward rate, $\mathrm{C}(s)$ is the cognitive cost of loading a new task frame, $\mathrm{E}(s)$ is environmental reinforcement, and $\alpha, \beta, \gamma > 0$ weight the agent's sensitivity to each component.

*Remark* 2.3. This definition is useful because it separates three business-relevant levers. If a person repeatedly fails to act, the intervention might target the reward gradient, the cognitive startup cost, or the environment. Treating all failure as "motivation" hides that design choice.

## 2.2 Target States and Barriers

**Definition 2.4** (Target state)**.** Given a current state $s_0$, a target state $s_1$ is a behavioural state whose expected long-run value exceeds that of $s_0$ under the agent's stated goals.

**Definition 2.5** (Activation barrier)**.** The activation barrier for transition $s_0 \to s_1$ is denoted by $\Delta V(s_0 \to s_1)$ and represents the minimal effort needed to interrupt the current frame and stabilize the new one.

**Proposition 2.6** (Barrier decomposition)**.** *For practical planning, the barrier can be decomposed as*

$$\Delta V(s_0 \to s_1) \approx b_0 + \eta_1 \mathrm{FI}(s_0) + \eta_2 U(s_1) - \eta_3 P,$$

*where $U(s_1)$ is startup ambiguity in the target state and $P$ is prior preparation.*

**Discussion**

The proposition is intentionally operational. It says that high inertia in the present, vagueness in the future, and lack of preparation all push in the same direction: they raise the threshold for action. The only term with the opposite sign is preparation. That is why planning matters even before one speaks about willpower.

## 2.3 Worked Example: Bed to Desk

**Example 2.7** (A minimal state transition)**.** Consider the transition from scrolling in bed to seated reading at a desk.

1. $\mathrm{R}(s_0)$ is high because scrolling offers rapid novelty.
2. $\mathrm{C}(s_0)$ is high because the reader must remember where the chapter, notebook, and task entry point are.
3. $\mathrm{E}(s_0)$ is high because posture and location both stabilize rest.
4. $U(s_1)$ is high if the study task is not pre-specified.

Packing the desk, opening the book in advance, and writing a first micro-task all raise $P$ and shrink $\Delta V$ before the transition even begins.

## 2.4 Notation as a Management Tool

Notation is not cosmetic. It serves two management functions:

1. it turns vague complaints into diagnosable variables;
2. it lets multiple writers and reviewers talk about the same failure pattern without renaming the problem each time.

**Chapter Summary**

The first chapter turns the manuscript from rhetoric into a controllable language. Once state, inertia, and barrier are named, later chapters can ask sharper questions: where is the barrier high, when does it drop, and which inputs change it most cheaply?

# Chapter 3

# Dynamics and Decision Windows

## 3.1 Volitional Energy

**Definition 3.1** (Volitional energy)**.** Volitional energy $\mathrm{W}(t)$ is the limited capacity available at time $t$ to interrupt the current behavioural frame and initiate stabilizing action toward a target frame.

**Definition 3.2** (Leverage)**.** The effective leverage of free will at time $t$ is

$$\mathrm{L}(t) = \frac{\mathrm{W}(t)}{\Delta V(t)}.$$

**Theorem 3.3** (Decision-window principle)**.** *Free will has maximum strategic effect not when effort is most intense, but when the activation barrier is locally minimal relative to available volitional energy.*

*Proof sketch.* If leverage is represented by $\mathrm{W}(t)/\Delta V(t)$, then leverage rises when $\mathrm{W}(t)$ increases or $\Delta V(t)$ decreases. In ordinary life, large increases in available will are unreliable, but temporary drops in barrier are common: after waking, after finishing a task, upon entering a new environment, or immediately after a sharp emotional update. Therefore the most reliable strategy is to act during barrier troughs rather than waiting for heroic motivation. □

## 3.2 Decision Windows

**Definition 3.4** (Decision window)**.** A decision window is a short interval during which the current state has lost part of its stabilizing force while the next state has not yet been rejected.

Common windows include:

- waking before distraction is loaded;
- the minute after finishing a scheduled block;
- physical arrival at a task-specific location;
- social commitments that force a new frame to begin.

## 3.3 Failure Modes

**Lemma 3.5** (Late-response failure)**.** *If the agent waits until inertia has fully rebuilt after a window closes, the same target action requires strictly greater effort than during the window.*

**Discussion**

This lemma explains a common misunderstanding. People often believe they have "decided" to act because they endorsed a future action verbally. But if they do not move inside the decision window, the state soon becomes expensive again and the earlier verbal decision has almost no mechanical value.

## 3.4 A Comparative Map of Windows

| Window type | Typical duration | Strategic use |
|---|---|---|
| Morning reset | 5–20 minutes | Inject a pre-written plan before default inputs arrive |
| Task boundary | 1–5 minutes | Move directly into the next prepared task while switching cost is already being paid |
| Spatial transition | Variable | Attach target behaviour to places that make old cues less available |
| Social trigger | Variable | Convert personal intention into a scheduled external obligation |

**Chapter Summary**

The second chapter explains why raw effort is a poor operating principle. Timing dominates intensity. The business value of the model comes from locating repeatable low-barrier windows and treating them as scarce assets.

# Chapter 4

# Planning as an Architectural Control

## 4.1 Planning Before the Moment of Action

**Theorem 4.1** (Planning theorem)**.** *Preparation performed during a high-clarity period lowers future activation barriers more reliably than last-minute self-persuasion.*

**Discussion**

Planning works on at least four channels:

1. it removes ambiguity from the target state;
2. it shortens the chain of in-the-moment decisions;
3. it places material cues into the environment;
4. it can import external force through schedules, reminders, or other people.

As a result, planning is not merely motivational language. It is a structural rewrite of future choice conditions.

## 4.2 A Planning Stack

For teaching purposes, it is useful to separate planning into layers:

1. **Outcome layer:** what end state is desired?
2. **Session layer:** what concrete session will move that outcome?
3. **Entry layer:** what exact first action starts the session?
4. **Environment layer:** what physical arrangement lowers entry friction?
5. **Trigger layer:** when and where is the decision window expected?

**Proposition 4.2** (Binary execution advantage)**.** *When a plan reduces a future action to a binary choice—execute or refuse—it consumes less volitional energy than a plan that still requires on-the-spot design.*

## 4.3 Chapter-Level Workflow

**Template**

**Minimal transition plan**

1. Name the target state.
2. Name the likely decision window.
3. Write the first visible action.
4. Place the required objects in the future environment.
5. Define the stop rule for the session.

## 4.4 Why Unplanned Intentions Fail

**Corollary 4.3** (The unplanned agent follows current inertia)**.** *If no preparation lowers future barrier terms, then the default winner of the next decision is the present state rather than the verbally preferred state.*

**Example 4.4** (Exercise after work)**.** The sentence "I should work out after work" is not yet a usable plan. It leaves open the location, bag, sequence, exercise selection, time length, and exit condition. Those unanswered questions are hidden barrier terms. A real plan resolves them upstream.

**Chapter Summary**

This chapter turns planning into infrastructure. Its value for the project is that it gives later case studies and exercises a stable template: identify the window, lower ambiguity, preload the environment, and collapse future choice.

# Chapter 5

# Case Library

## 5.1 Why the Book Needs Cases

Without a case library, the model risks sounding universally true while remaining operationally thin. Cases reveal whether the same language explains different behavioural failures without being stretched beyond recognition.

## 5.2 Case 1: Study Start Failure

### Situation

A student returns from lunch intending to read a difficult chapter but instead opens short-video feeds for forty minutes.

### State analysis

- Current reward is high because novelty and low effort combine.
- Switching cost is high because the next theorem proof requires context reload.
- Environmental reinforcement is high because the phone is in hand and the desk is not prepared.

### Interpretation

The failure is not primarily lack of educational value alignment. It is a barrier geometry problem. The reader entered the post-lunch window without a prepared target state.

### Repair

Write the next proof target before the meal break, leave the relevant page open, and move the phone physically out of the work zone. These steps lower C and E before the window arrives.

## 5.3 Case 2: Exercise Non-Start After Work

### Situation

An employee endorses regular exercise every morning but rarely reaches the gym after a draining workday.

### State analysis

- W is low by evening.
- $\Delta V$ is inflated by route uncertainty, clothing friction, and social isolation.
- Home arrival rebuilds a rest-oriented environment too quickly.

### Interpretation

The problem is not failure to choose health in the abstract. The problem is that the critical transition happens at the worst point in the daily cycle.

### Repair

Shift the window upstream: pack the bag in advance, choose one default workout, and route the commute through the gym. The intervention works because it moves the decision from "Should I engineer a workout now?" to "Do I enter the door I already planned to enter?"

## 5.4 Case 3: Writing Avoidance in Research Work

### Situation

A graduate student can read papers and discuss ideas but repeatedly delays writing drafts.

### State analysis

- The reward gradient favors research browsing because it feels productive without exposing unclear prose.
- The target state has a large ambiguity term: what section, what claim, what sentence?
- The current state protects self-image because it avoids visible output.

### Interpretation

This is a form of pseudo-work inertia. The agent is not inactive; the agent is trapped in a neighbouring state whose short-run reward is safer than draft production.

### Repair

Define the next writing session as "produce five ugly bullet claims for Section 2." The target state becomes concrete enough to enter. Quality can be optimized later; entry cannot.

## 5.5 Case 4: Sleep Schedule Breakdown

### Situation

A person wants to sleep earlier but continues consuming stimulating content at night until bedtime shifts later and later.

### State analysis

- Night content has high immediate reward and low resistance.
- The target state "go to sleep" has weak action identity because it is treated as absence of stimulation rather than a designed transition.
- The environment keeps bright screens and restless cues active.

### Interpretation

The agent is trying to defeat a high-reward state using only a negative rule: "stop." That rule does not specify a replacement frame.

### Repair

Design a pre-sleep state with a location, object set, and first action: dim light, paper book, charger placed away from bed, and a fixed shutdown cue. The target becomes a positive state rather than an empty prohibition.

## 5.6 Cross-Case Comparison

| Case | Dominant barrier | Missed window | Best repair |
|---|---|---|---|
| Study start | Switching cost | Post-break re-entry | Preload the exact next problem |
| Exercise start | Low evening will | Commute-to-home boundary | Shift the route and pre-pack gear |
| Writing avoidance | Target ambiguity | First writing minute | Shrink the target to ugly visible output |
| Sleep repair | Rewarded default state | Night shutdown transition | Build a positive replacement ritual |

**Chapter Summary**

The case library increases transfer value. The model now explains not one habit, but several structurally different failures. That makes the framework more credible as a textbook system and gives later readers reusable pattern recognition rather than isolated anecdotes.

# Chapter 6

# Intervention Design Patterns

## 6.1 From Diagnosis to Design

Once the current failure is classified, intervention design should ask a simple question: which term in the barrier is cheapest to change first? In practice, cheap interventions usually beat heroic ones.

## 6.2 Comparative Intervention Table

| Intervention family | Primary target | When to use | Typical action |
|---|---|---|---|
| Preparation | $\Delta V$ and ambiguity | Target is vague | Pre-write the first visible step |
| Environmental redesign | E | Cues keep restoring the old state | Remove distractors or relocate entry point |
| Sequencing | C | Transition cost is high | Attach action to a pre-existing boundary |
| External commitment | effective force | Private resolve is weak | Add another person, schedule, or deadline |
| Minimum viable entry | perceived barrier | Fear of scale blocks start | Define a tiny but real first session |

## 6.3 Design Principles

**Proposition 6.1** (Cheapest-term principle)**.** *When several barrier terms are elevated, the first intervention should usually target the cheapest high-impact term rather than the most philosophically important term.*

**Discussion**

Suppose distraction, ambiguity, and fatigue are all present. Asking the agent to "want success more" is expensive and unreliable. In contrast, moving the phone away and naming the first task are cheap and immediate. Textbook design should privilege the cheapest controllable variable.

## 6.4 Implementation Template

**Template**

**Framework Inertia intervention sheet**

1. Current state:
2. Target state:
3. Dominant barrier term (R / C / E / ambiguity / timing):
4. Decision window to exploit:
5. One environmental change:
6. One preparation change:
7. One minimum viable entry action:
8. Verification rule after the session:

## 6.5 Worked Implementation

**Example 6.2** (Turning a reading intention into a design)**.** Suppose the target is daily technical reading.

1. Current state: post-dinner phone browsing.
2. Target state: twenty minutes of annotated reading.
3. Dominant barrier: high ambiguity and high phone reward.
4. Window: immediately after dinner cleanup.
5. Environmental change: place the phone in another room.
6. Preparation change: leave the book open with one written guiding question.
7. Minimum viable entry: read one page and write one margin note.
8. Verification: log whether the first page was completed before 8:15 PM.

This design succeeds because it changes multiple barrier terms before the window closes and gives the reader a concrete completion rule.

## 6.6 Failure Review Protocol

When an intervention fails, the review should be diagnostic rather than moral:

1. Did the decision window actually occur?
2. Was the first action visible enough?
3. Did the environment still support the old state?
4. Was the target state too ambiguous?

5. Was the session too large for the available window?

**Chapter Summary**

This chapter makes the theory commercially useful. It converts the model from explanation into a repeatable operating pattern: diagnose the dominant term, choose the cheapest meaningful intervention, and review failure without losing the structure of the model.

# Chapter 7

# Exercises and Reflection Prompts

## 7.1 Short Answer Questions

**Exercise Set**

**Exercise 7.1.** State the difference between framework inertia and volitional energy in one paragraph. Why is it a mistake to collapse both into the single word "motivation"?
**Exercise 7.2.** Give one example of a decision window that occurs in your own daily routine. Explain why the window is short-lived.
**Exercise 7.3.** Describe a case in which the target state was too ambiguous to enter. Which term in the barrier decomposition captured that ambiguity?

## 7.2 Applied Modelling Tasks

**Exercise Set**

**Exercise 7.4.** Model the transition from passive information consumption to active note-taking. Name the current state, target state, dominant reward source, switching cost, and one environmental reinforcement.
**Exercise 7.5.** Choose a repeated behaviour failure and design a one-page intervention using the template from Chapter 6. Your answer must include a decision window and a verification rule.

## 7.3 Contrast and Boundary Questions

**Exercise Set**

**Exercise 7.6.** Give an example of a case where the main problem is not high current inertia but low clarity about long-run goals. Explain why the framework is still useful, but not sufficient by itself.
**Exercise 7.7.** When might an external commitment lower barriers for one person but raise them for another? Answer using the language of reward, environment, and cognitive cost.

## 7.4 Self-Audit Worksheet

**Template**

**Weekly framework inertia audit**

1. What beneficial transition did I fail to start this week?
2. What was the current state that won instead?
3. Which barrier term dominated the failure?
4. Which decision window did I miss?
5. What one preparation change will I install before the next attempt?
6. What evidence next week would count as real improvement?

## 7.5 Instructor-Oriented Prompting Uses

This chapter can also support discussion-based teaching:

- ask students to compare two failed transitions with the same model;
- ask for a case where the best intervention is environmental rather than motivational;
- ask students to redesign a failure using a smaller entry action.

**Chapter Summary**

The exercise layer turns the manuscript from a reading artifact into a usable training artifact. That matters because a textbook without exercises cannot prove that its concepts transfer beyond the page.

# Appendix A

# Notation and Planning Templates

## A.1 Notation Table

| Symbol | Meaning |
|---|---|
| $s$ | behavioural state |
| $\mathrm{FI}(s)$ | framework inertia of state $s$ |
| $\mathrm{R}(s)$ | immediate reward rate of the current state |
| $\mathrm{C}(s)$ | cognitive switching cost of leaving the current state |
| $\mathrm{E}(s)$ | environmental reinforcement of the current state |
| $\Delta V(s_0 \to s_1)$ | activation barrier for a transition |
| $\mathrm{W}(t)$ | volitional energy available at time $t$ |
| $\mathrm{L}(t)$ | effective leverage of free will at time $t$ |
| $P$ | prior preparation that lowers future barrier terms |

## A.2 Planning Templates

**Template**

**Session-design card**

1. Target outcome for the week:
2. Next session that serves the outcome:
3. Exact first action:
4. Objects that must already be placed:
5. Window in which the session should begin:
6. Rule for ending the session:

**Template**

**Failure review card**

1. Which state won?
2. Which barrier term was underestimated?
3. Which cue restored the old state?
4. Which smaller entry action should replace the current one?
5. What will be changed before the next window arrives?

## A.3 Appendix Use

The appendix exists for reuse. If later versions add more formal proofs, empirical citations, or domain-specific case banks, those modules can still share the same notation and planning templates. That keeps the manuscript coherent as it grows.